

3.5 Carbonyl Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.5 Carbonyl Compounds
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

What is formed when butanone is refluxed with a solution of NaBH_4 ?

- A. Butane
- B. Butan-1-ol
- C. Butan-2-ol
- D. Butanal

[1 mark]

Question 2

In 1903 Arthur Lapworth became the first chemist to investigate a reaction mechanism. The reaction of hydrogen cyanide with propanone.

What do we now call the mechanism of this reaction?

- A. Nucleophilic substitution
- B. Nucleophilic addition
- C. Electrophilic substitution
- D. Electrophilic addition

[1 mark]

Question 3

Which carbonyl compound(s) react with **both** LiAlH_4 and Tollens' reagent?

- A. Ketones only
- B. Aldehydes only
- C. Both aldehydes and ketones
- D. Neither aldehydes nor ketones

[1 mark]

Question 4

In a nucleophilic addition reaction, aqueous methanolic solution of NaBH_4 acts as a reducing agent when added to propanal, $\text{CH}_3\text{CH}_2\text{CHO}$.

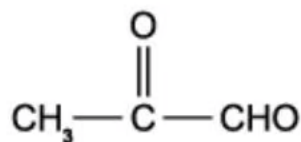
What is the first step of this mechanism?

- A. Attack of an H^- ion at the oxygen atom of the carbonyl group.
- B. Attack of an H^+ ion at the oxygen atom of the carbonyl group.
- C. Attack of an H^- ion at the carbon atom of the carbonyl group.
- D. Attack of an H^+ ion at the carbon atom of the carbonyl group.

[1 mark]

Question 5

The characteristic smell of burnt sugar is partly caused by the following compound.



There are two functional groups in the compound.

Which reagent will react with **both** functional groups?

- A. Sodium hydroxide
- B. Hydrogen cyanide
- C. Fehling's solution
- D. Acidified potassium dichromate (VI)

[1 mark]

Question 6

In which reaction is the organic compound oxidised?

- A. $\text{CH}_3\text{CH}_2\text{CN} + \text{dilute H}_2\text{SO}_4$
- B. $\text{CH}_3\text{CH}_2\text{CHO} + \text{Tollens' reagent}$
- C. $\text{CH}_3\text{COCH}_2\text{CH}_3 + 2,4\text{-dinitrophenylhydrazine reagent}$
- D. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{concentrated H}_3\text{PO}_4$

[1 mark]

Question 7

Ethanal, CH_3CHO , is reduced using NaBH_4 in aqueous ethanol in a nucleophilic addition reaction.

What is the first step of this mechanism?

- A. The oxygen atom of the carbonyl group is attacked by an H^- ion
- B. The oxygen atom of the carbonyl group is attacked by an H^+ ion
- C. The carbon atom of the carbonyl group is attacked by an H^- ion
- D. The carbon atom of the carbonyl group is attacked by an H^+ ion

[1 mark]

Question 8

The compounds below are treated with hydrogen cyanide.

- 1 Butanal
- 2 Pentan-3-one
- 3 2-chlorobutane

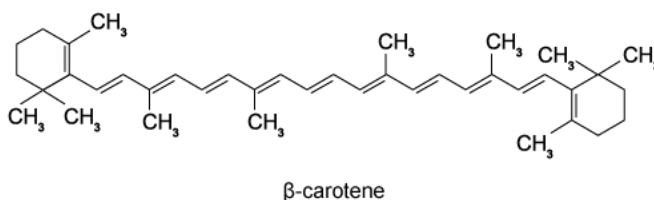
Which compounds react and produce a molecule containing a chiral centre?

- A. 1 only
- B. 1 and 2
- C. 2 only
- D. 2 and 3

[1 mark]

Question 9

The orange colour of carrots is due to the presence of β -carotene, the structure of which is shown in the diagram below.



Hot, concentrated, acidified KMnO_4 will oxidise β -carotene.

When an individual molecule of β -carotene is oxidised in this way there are multiple possible products.

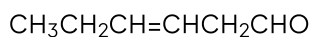
How many of these product molecules contain a ketone functional group?

- A. 3
- B. 6
- C. 9
- D. 12

[1 mark]

Question 10

Hex-3-enal is used as a flavouring agent.



hex-3-enal

Which row correctly shows the product formed when hex-3-enal is treated with the given reducing agent?

	Product	Reducing agent
A	$\text{CH}_3(\text{CH}_2)_4\text{CH}_3$	$\text{H}_2 + \text{Ni}$
B	$\text{CH}_3(\text{CH}_2)_4\text{CH}_2\text{OH}$	NaBH_4
C	$\text{CH}_3(\text{CH}_2)_4\text{CHO}$	$\text{H}_2 + \text{Ni}$
D	$\text{CH}_3(\text{CH}_2)_4\text{CHO}$	NaBH_4

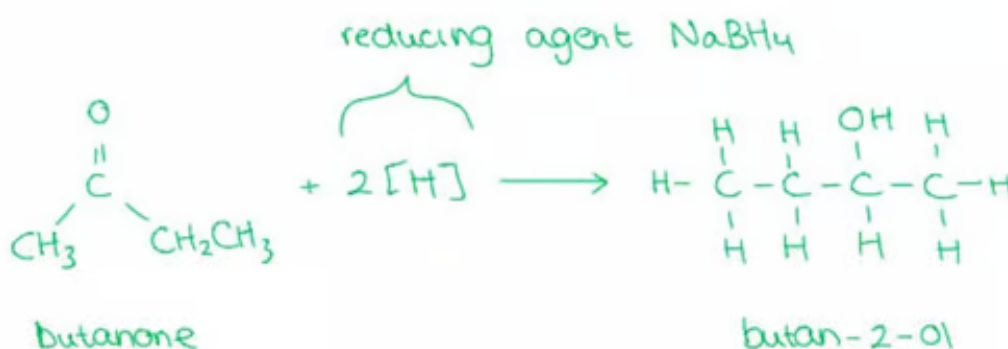
[1 mark]

Model Answers: Easy

1

The correct answer is **C** because:

- **Reduction** is the addition of a hydrogen to an organic compound.
- **Butanone** is a **ketone** and ketones undergo reduction reactions when **refluxed** with a **reducing agent** such as **NaBH₄** or **LiAlH₄**.
- A **secondary alcohol** is formed from the reduction of a ketone.
 - **Butan-2-ol** is a secondary alcohol.
- The reaction mechanism for this reaction is:



A is incorrect as butane is an **alkane**. The reduction of a ketone with NaBH₄ produces a **secondary alcohol**.

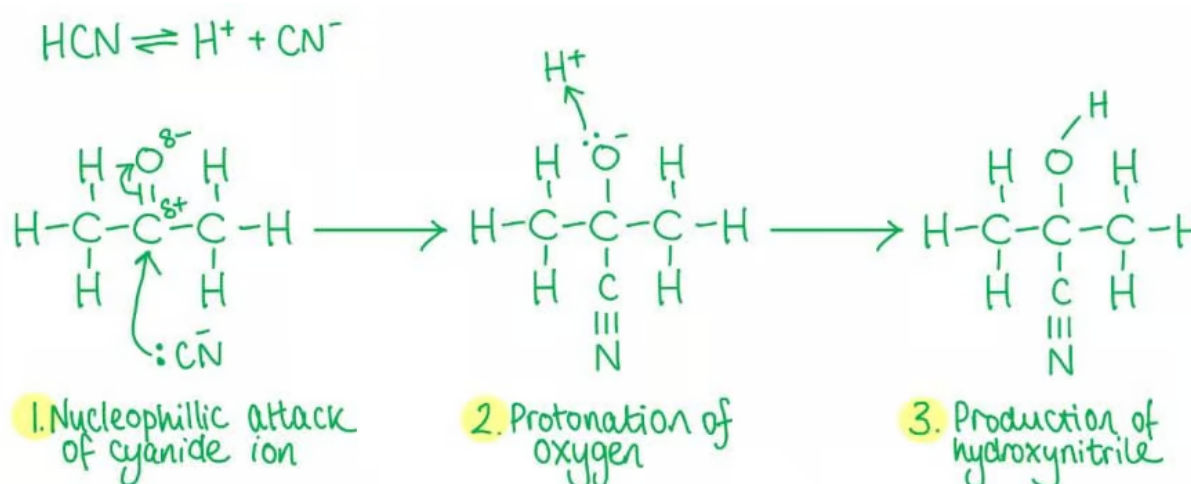
B is incorrect as butan-1-ol is a primary alcohol and would have been formed from the reduction of an aldehyde (butanal) with NaBH₄.

D is incorrect as butanal is an aldehyde. The reduction of a ketone with NaBH₄ produces a secondary alcohol.

2

The correct answer is **B** because:

- A **nucleophilic addition** is when a **nucleophile** is added to an unsaturated bond to form a **single** product.
 - A nucleophile is a species that donates a pair of electrons.
- **Hydroxynitriles** can be formed from the **nucleophilic addition** reaction of aldehydes and ketones with HCN, KCN as catalyst and heat.
- The cyanide ion, CN^- , acts as a **nucleophile** and is added on to the carbonyl carbon to form a single product only.
- The reaction of **propanone** (which is a **ketone**) with **hydrogen cyanide** (which is the nucleophile) will result in the formation of 2-methyl-2-hydroxybutanenitrile.
- The reaction mechanism for this reaction is:



A is incorrect as the cyanide ion, CN^- , is a **nucleophile** but it is not **replacing** another atom or atom group in **propanone**. The reaction only produces one product (which is characteristic of an **addition** reaction) and does not produce two products (which is characteristic of a **substitution** reaction).

C & D are incorrect as the cyanide ion, CN^- , is a **nucleophile** and not an **electrophile**. The reaction that takes place is not a **substitution** but an **addition** reaction as only one product is formed.

3

The correct answer is **B** because:

- Tollens' reagent is an aqueous solution of **silver nitrate ions** in **excess ammonia**

solution used to test for the presence of **aldehyde** compounds.

- The silver ions, **Ag⁺**, in the solution act as mild **oxidising agent**.
 - When warmed with the silver ions, an **aldehyde** will be **oxidised** to form a **carboxylic acid** which in turn is **neutralised** to the **carboxylate ion**, -COO^- , as H^+ is removed from the -COOH in the alkaline conditions of the test and a salt is formed.
 - The silver ions themselves are **reduced** to **silver atoms**.
 - The silver atoms form a '**mirror**' on the inside of the tube giving a positive test for an aldehyde.
 - **No change is observed** when a ketone is warmed in Tollens' reagent as no **redox reaction** can take place causing the mixture in the test-tube to remain **colourless**.
- **LiAlH₄** is a **strong reducing agent** reducing both aldehydes and ketones.
 - A reducing agent is a species that **reduces** another species and gets **oxidised** itself.
 - Aldehydes are reduced to **primary alcohols**.
 - Ketones are reduced to **secondary alcohols**.

A & C are incorrect as ketones do **not** react with Tollens' reagent as they cannot be further oxidised.

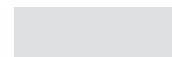
D is incorrect as both aldehydes and ketones react with **LiAlH₄**. Only aldehydes react with Tollens' reagent.

4

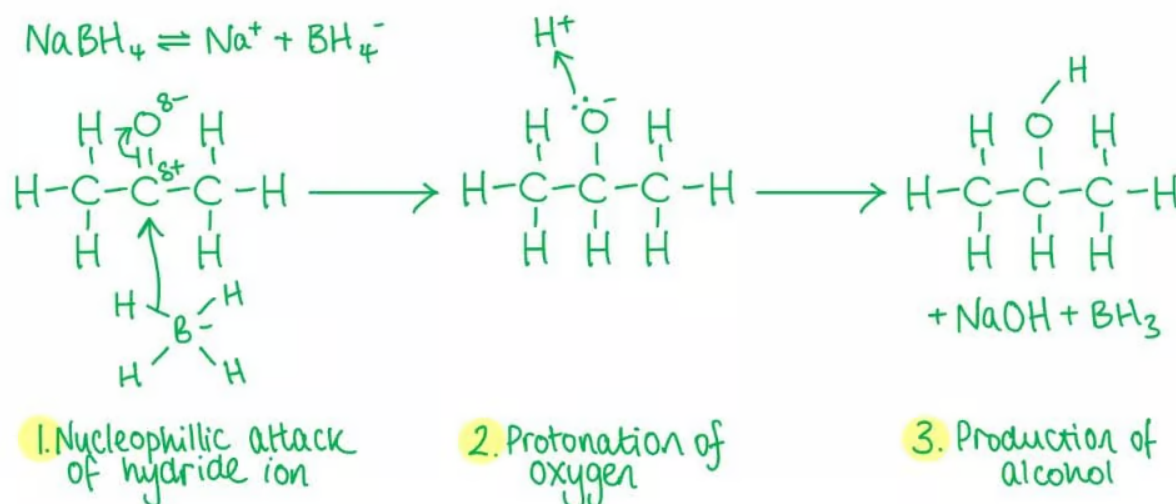
The correct answer is **C** because:

- **Reduction** is the addition of a hydrogen to an organic compound.
- **NaBH₄** is a **reducing agent**.
- Aldehydes are **reduced** to **primary alcohols** by warming them with an aqueous alkaline solution of NaBH₄.
- Ketones are reduced to **secondary alcohols** by warming them with an aqueous alkaline solution of NaBH₄.

- The carbonyl group in aldehydes and ketones is polarised due to high electronegativity of the oxygen atom.
- This makes the carbon C atom open to attack by a **nucleophile** (in this case H^-).
- The **negatively charged intermediate** formed in the first step in the mechanism is very reactive and reacts quickly with an aqueous H^+ ion to form an **alcohol**.
- The reaction mechanism for this reaction is:



- The reaction mechanism for this reaction is:



A is incorrect as the H^- is a **nucleophile** and will attack a partial positively charged centre. The oxygen atom of the carbonyl group is a partial negatively charged centre and will repel the H^- ion.

B & D are incorrect as propanal reacts with an **alkaline** (aqueous methanolic solution) of NaBH_4 and not an **acidic** solution of NaBH_4 .

5

The correct answer is **B** because:

- Hydroxynitriles** can be formed from the **nucleophilic addition** reaction of **aldehydes** and **ketones** with HCN , KCN as catalyst and heat.
- The cyanide ion, CN^- , acts as a **nucleophile** and is added on to the carbonyl carbon to form a single product only.
- Both the **aldehyde** and **ketone** in the compound in the question will undergo a **nucleophilic addition** reaction with HCN .

A is incorrect as **aldehydes** and **ketones** don't react with **sodium hydroxide**.

C is incorrect as the Fehling's solution is an **alkaline** solution containing copper (II) ions. When warmed with an **aldehyde**, the Cu^{2+} are reduced to Cu^+ changing the clear **blue** Fehling's (CuSO_4) solution colour to an opaque **red precipitate** of copper (I) oxide forms. Fehling's solution will therefore only react with the aldehyde part of the molecule and not the ketone group.

D is incorrect as **acidified potassium dichromate (VI)** is an **orange** oxidising agent. The reaction mixture turns **green** as the orange dichromate ions ($\text{Cr}_2\text{O}_7^{2-}$) are reduced to **green** Cr^{3+} ions. Primary alcohols, secondary alcohols and aldehydes can be oxidised by acidified potassium dichromate (VI) the ketone group of the compound cannot be oxidised by acidified potassium dichromate (VI).

6

The correct answer is **B** because:

- Tollens' reagent is an aqueous solution of **silver nitrate** in **excess ammonia solution** used to test for the presence of **aldehyde** compounds.
- The silver ions, **Ag^+** , in the solution act as mild **oxidising agent**.
- When warmed with the silver ions, an **aldehyde** will be **oxidised** to form a **carboxylic acid** which in turn is **neutralised** to the **carboxylate ion**, -COO^- , as H^+ is removed from the -COOH in the alkaline conditions of the test and a salt is formed.
- The silver ions themselves are **reduced** to **silver atoms**.
- The silver atoms form a '**mirror**' on the inside of the tube giving a positive test for an aldehyde.
- $\text{CH}_3\text{CH}_2\text{CHO}$ is an **aldehyde** and will react with the Tollens' reagent to form $\text{CH}_3\text{CH}_2\text{COOH}$ (propanoic acid).

A is incorrect as this is an example of a **hydrolysis** reaction of nitriles with **dilute acids** (such as sulfuric acid in this example). The products of this hydrolysis are propanoic acid and an ammonium salt.

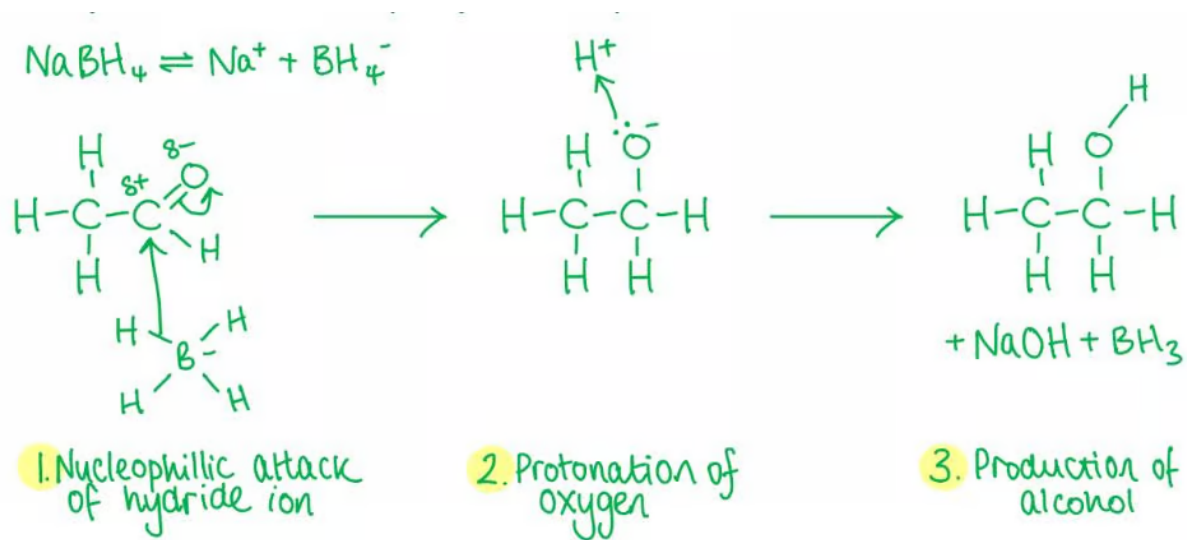
C is incorrect as the presence of a **carbonyl group (C=O)** can be tested by adding a solution of 2,4-dinitrophenylhydrazine (**2,4-DNPH**) which is an orange liquid – if an aldehyde or ketone is present, a **deep-orange precipitate** is formed. This reaction is an example of a **condensation reaction** in which two molecules join together and **eliminate a small molecule**.

D is incorrect as alcohols react with concentrated phosphoric acid to generate an **alkene**. The alcohol **loses** an oxygen and is therefore **reduced** and not **oxidised**. This is also called the dehydration of alcohol (a water molecule is removed from the alcohol).

7

The correct answer is **C** because:

- **NaBH_4 is a reducing agent.**
 - A reducing agent is a species that **reduces** another species and gets **oxidised** itself.
- Aldehydes are **reduced** to **primary alcohols** by warming them with an aqueous alkaline solution of NaBH_4 .
- Ketones are reduced to **secondary alcohols** by warming them with an aqueous alkaline solution of NaBH_4 .
- The carbonyl group in aldehydes and ketones is **polarised** due to high **electronegativity** of the oxygen atom.
- This makes the carbon C atom open to attack by a **nucleophile** (in this case H^-).
- The negatively charged intermediate formed in the first step in the mechanism is very reactive and reacts quickly with an aqueous H^+ ion to form an **alcohol**.



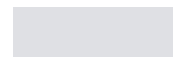
A is incorrect as the H^- is a **nucleophile** and will attack a partial positively charged centre. The oxygen atom of the carbonyl group is a partial negatively charged centre and will repel the H^- .

B & D are incorrect as the propane reacts with an alkaline (aqueous methanolic solution) of NaBH_4 and not an acidic solution of NaBH_4 .

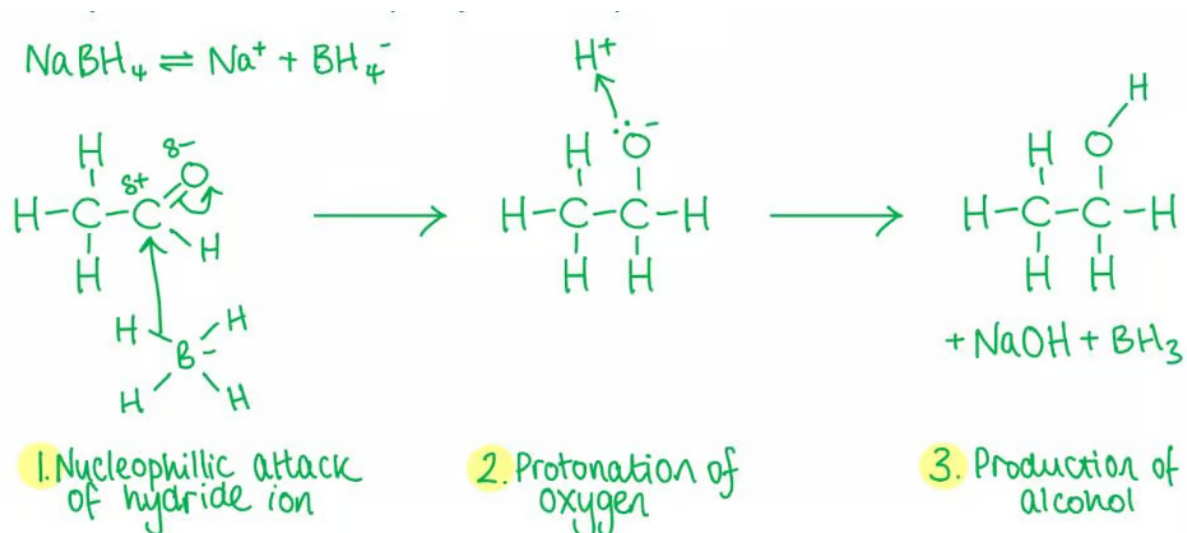
8

The correct answer is **A** because:

- **Hydroxynitriles** can be formed from the **nucleophilic addition** reaction of **aldehydes** and **ketones** with HCN, KCN as catalyst and heat.
- The cyanide ion, CN^- , acts as a **nucleophile** and is added on to the carbonyl carbon to form a single product only.



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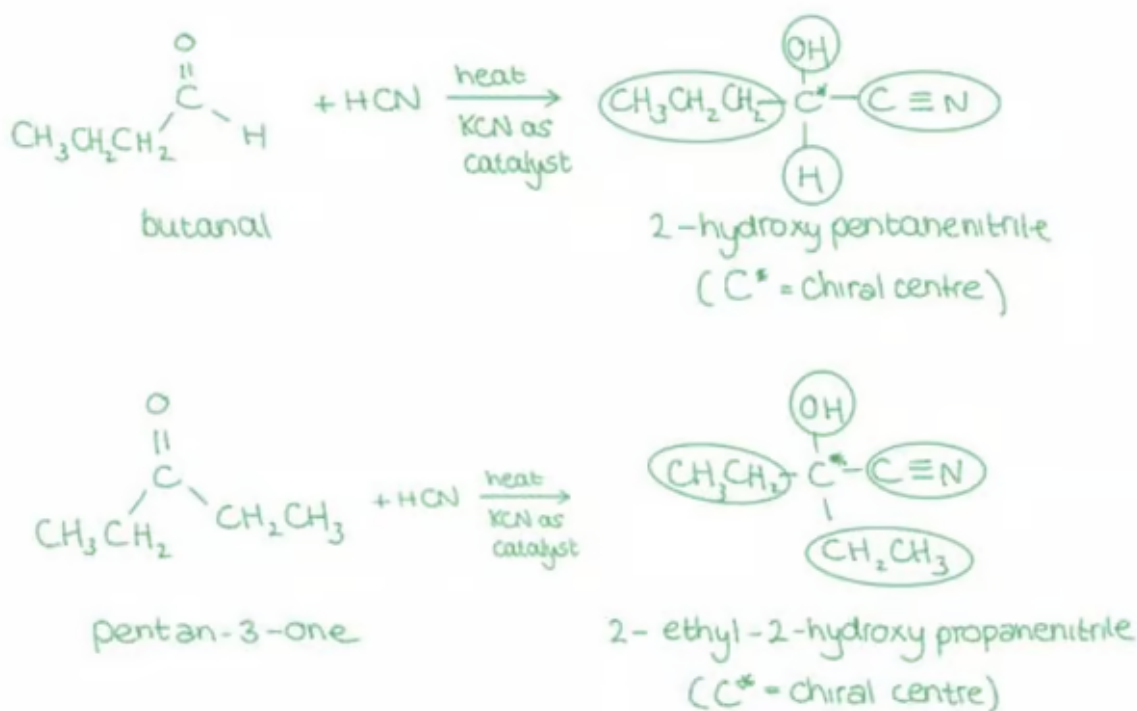
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B & D are incorrect as the propane reacts with an alkaline (aqueous methanolic solution) of NaBH_4 and not an acidic solution of NaBH_4 .

8

The correct answer is **A** because:

- **Hydroxynitriles** can be formed from the **nucleophilic addition** reaction of **aldehydes** and **ketones** with HCN , KCN as catalyst and heat.
- The cyanide ion, CN^- , acts as a **nucleophile** and is added on to the carbonyl carbon to form a single product only.
- The reaction of **butanal** and **pentan-3-one** with **hydrogen cyanide** (which is the **nucleophile**) will result in the formation of a **hydroxynitrile**.
- The hydroxynitrile produced has a chiral centre as the carbon attached to the nitrile and hydroxy group is bonded to four different **atoms** or **groups of atoms**.



B & C are incorrect as pentan-2-one does not give a product with a chiral centre.

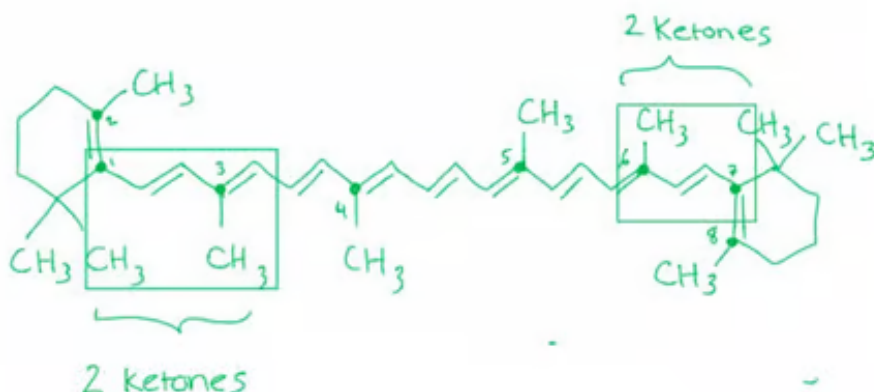
D is incorrect as the reaction of 2-chlorobutane with hydrogen cyanide is a **nucleophilic substitution** reaction. CN⁻ acts as a **nucleophile** and **replaces** the chlorine in 2-chlorobutane to form pentanenitrile. None of the carbons in pentanenitrile are bonded to four different atoms or atom groups therefore the product does not contain any **chiral centres**.

9

The correct answer is **B** because:

- **Oxidation** in organic chemistry is the gain of oxygen atoms.
- The acidified potassium manganate (VII) solution **oxidises** the **alkene** by **breaking** the C=C double bond and replacing it with two C-O double bonds.
- The products are known as **carbonyl compounds**.
- In β-carotene one should look at the carbon atoms in the double bonds which are attached to **two** carbon atoms.
 - There are 8 of these carbons which are attached to two other carbon atoms and which can form a ketone functional group.

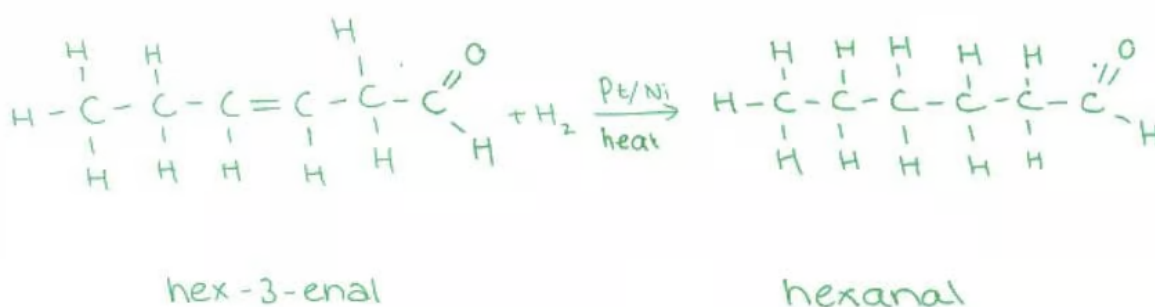
- Upon oxidation, the double bonds are broken forming 6 product molecules.
 - 4 of these product molecules contain one ketone group.
 - 2 of these product molecules contain 2 ketone groups.



10

The correct answer is **C** because:

- The two functional groups present in hex-3-enal are the **alkene** (C=C) and **aldehyde** (CHO) groups.
- **H₂** is a **reducing agent**.
 - A reducing agent is a species that **reduces** another species and gets **oxidised** itself.
- The reaction of alkenes with H₂ and nickel or platinum as a catalyst is a **hydrogenation** reaction.
 - A **hydrogenation** reaction is the addition of hydrogen to a double bond.
- The C=C double bond is reduced by the addition of H₂ while the carbonyl group remains unaffected.



A is incorrect as the reaction of an alkene with H_2 and Ni as a catalyst is a **hydrogenation reaction**. The hydrogen is added to the $\text{C}=\text{C}$ double bond while leaving the **carbonyl group unaffected**.

B & D are incorrect as aldehydes react with NaBH_4 and are reduced to **primary alcohols**. NaBH_4 is not a strong enough reducing agent to also reduce the alkene so the alkene will remain unaffected by the reaction of hex-3-enal with NaBH_4 .

